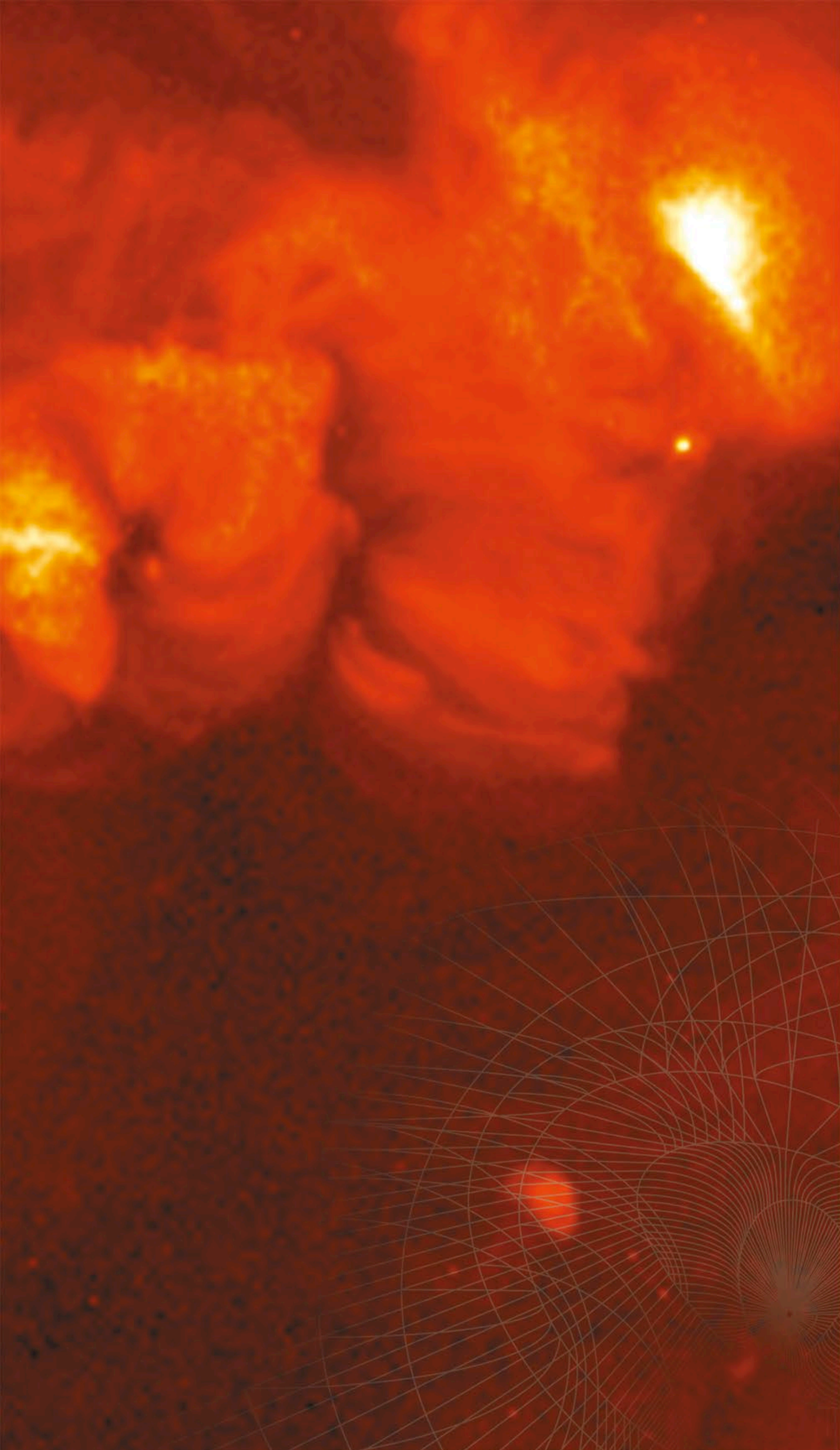
THE SOLAR SYSTEM

"How vast those Orbs must be, and how inconsiderable this Earth, the Theatre upon which all our mighty Designs, all our Navigations, and all our Wars are transacted, is when compared to them."

Christiaan Huygens

THE SOLAR SYSTEM IS the region of space that falls within the gravitational influence of the Sun, an ordinary yellow star that has shone steadily for almost 5 billion years. After the Sun itself, the most significant objects in the solar system are the planets—a group of assorted rocky, gaseous, and icy worlds that follow independent, roughly circular orbits around their central star. Most of the planets are orbited in turn by moons, while a huge number of smaller lumps of rock and ice also follow their own courses around the Sun—though largely confined in a few relatively crowded zones. Myriad tiny particles flow around all these larger bodies—ranging from fragments of atoms blown out by the Sun to motes of dust and ice left in the wake of comets. Our local corner of the universe has been studied intensively from the time of the first stargazers to the modern era of space probes, yet it is still a source of wonder and surprise.



SOLAR FLARE
On the broiling surface of the Sun, a cataclysmic release of magnetic energy triggers a solar flare—a violent outburst of radiation and high-energy particles that will reach Earth within hours.